# Internal Air Transport Certification

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**Date:** 7 July 2017
**Item Nomenclature:** FMTV A1P2 Series (Long Term Armor Strategy, LTAS): A-Cab and with B-kit (Armored Configuration)
**File Number:** 2008.01.13 Rev 6
**Requestor:** On File
**Superseded Certification Date:** 24 May 2017

**New Information Summary:** Added cargo bed tiedown provision rating.

**Reference Documents:**
1. Goodyear 395/85R20 LR-G MV/T Recommended Inflation Pressure for Aircraft Transport, 3 September 2009
2. Michelin North America 20 Jan 2003 memo, subject: Reduced Inflation Pressures for Aircraft Loading and Unloading.
3. Lockheed Martin, Rev 3 1 July 2009 LTS-0256 subject: Wheeled Vehicle Axle Load Limit Analysis
4. 330 ACSG/GFEA 12 August 2009 email, subject: FMTV
5. Director, Family of Medium Tactical vehicles BAE Systems Global Tactical Systems email, 21 October 2010, FMTV Axle Loading Approval (UNCLASSIFIED).

**Item Description:** The subject FMTV A1P2 series (LTAS) is identified to be upgraded FMTVs (A-cab) with an armored cab (B-kit). The vehicles are designed to carry a wide variety of cargo and weapons systems. See Tables 1, 2, and 3 for M-number, dimensions and ratings. The 2.5 ton Light Medium Tactical Vehicles (LMTV) has two axles. The 5 ton Medium Tactical Vehicles (MTV) has three axles.

**NOTE:** The listed dimensions are approximations. Actual dimensions can vary as long as aircraft limitations published in applicable cargo loading manuals and the conditions of certification listed below are not violated.




**Figure 1: Up-armored FMTVs**



**Figure 2: MTV with B-kit and Gunner Protection Kit (GPK)**

**Certified Aircraft:** USAF C-130 (all listed FMTV LTAS variants other than the M1089A1P2), C-17, and C-5

**Conditions of Certification:**

**1. Maximum Weight for Air Transport:**

a. C-130 transport:

(1) Gross Vehicle Weight:

(a) LMTV: 34,150 lbs

(b) MTV: 41,600 lbs

(2) Axle Limits:

(a) Front: 15,600 lbs (single axle limit per reference 4)

(b) Rear: LMTV - 15,000 lbs (LMTV single rear axle limit per reference 4)
MTV - 13,000 lbs each (Intermediate and aft axle - aircraft limitation)

b. C-17 and C-5:

(1) Gross Vehicle Weight: Listed GVWR in Tables 1, 2 and 3.

(2) Axle Limits: 19,000 lbs (without sleeper shoring); 22,700 lbs per reference 5 (with sleeper shoring)

c. Weight Limit for Secondary/Accompanying load on vehicle bed: 10,000 lbs

**2. Certified Vehicles:**

a. The vehicle/load combination can be transported provided it does not exceed size limits specified in the aircraft operating manual, does not exceed the FMTV weight limits citied in this airlift certification and attachment to the vehicle meets airlift restraint levels. Items certified for rail transport are acceptable for airlift. Additional restraint to the vehicle or aircraft floor may be placed on the accompanying load at the discretion of the loadmaster.

b. Notes:

(1) FMTV variants are 96” wide at the vehicle body.
(2) Winch installation will not increase overall length.

(3) Gun ring mounts increase overall height by approximately 3”.

(4) Gunner Protection Kit (GPK, turret from M1114) height is approximately 143”.

**Table 1: LMTV (2-1/2 Ton) Characteristics**

| **LMTV** (Single Rear Axle) | Length (in) | Height (in) (w/o load) | Front Axle Rating (lbs) | Rear Axle Rating (lbs) | GVWR (lbs) |
|---|---|---|---|---|---|
| M1078A1P2 | 264 | 110 | 19,000 | 15,000 | 34,150 |
| M1079A1P2 | 264 | 113/139 with van body | 19,000 | 15,000 | 34,150 |

**Table 2: MTV (5 Ton) Characteristics**

| **MTV** (Dual Rear Axle) | Length (in) | Height (in) | Front Axle Rating (lbs) | Rear Bogie Rating (lbs) | GVWR (lbs) |
|---|---|---|---|---|---|
| M1083A1P2 | 285 | 113 | 19,000 | 38,000 | 44,952 |
| M1084A1P2 | 306 | 113 | 19,000 | 38,000 | 44,952 |
| M1085A1P2 | 349 | 113 | 19,000 | 38,000 | 44,952 |
| M1086A1P2 | 394 | 113 | 19,000 | 38,000 | 44,952 |
| M1087A1P2 | 406 | 113/ 149 with van body | 19,000 | 38,000 | 44,952 |
| M1088A1P2 | 283 | 113 | 19,000 | 38,000 | 44,952 |
| M1089A1P2 | 376 | 121 | 19,000 | 38,000 | 44,952 |
| M1092A1P2 | 285 | 113 | 19,000 | 38,000 | 44,952 |
| M1157A1P2 | 300 | 113 | 19,000 | 38,000 | 54,890 |

**Table 3: MTV LHS Characteristics**

| **MTV LHS** | Length (in) | Width (in) | Height (in) | Front Axle Rating (lbs) | Rear Bogie Rating (lbs) | GVWR (lbs) |
|---|---|---|---|---|---|---|
| M1148A1P2 | 361-379 depending on payload | 97.5-100.5 depending on payload | 153-165 depending on payload | 19,000 | 38,000 | 50,888 |

**3. Item Preparation:**

a. Remove weapons from roof-mounted gun rings.

b. Remove and palletize payloads with heights exceeding aircraft capability.

c. C-130:

(1) The height of the FMTV vehicles, with accompanying load, will have to be reduced from its operational configuration to a maximum height of 106 inches or less. Equipment including the GPK, will have to be removed to reduce height. The tire inflation pressure shall be set to 10-12 psi for Goodyear tires, and 15 psi for Michelin tires in order to reduce the height of the vehicle.

(a) Data has been made available by the tire manufacturers, Goodyear and Michelin, validating that the vehicles are loadable onto the C-130 aircraft at this low pressure.

(b) To reduce the risk of the vehicle bouncing and contacting the ceiling of the C-130 sleeper shoring is required at all times.

(2) Per reference 4, the 330 ACSG/GFEA has authorized the loading and air transport of the subject vehicles at a maximum front axle weight of 15,600 lbs for C-130E/H/J and J-30 aircraft. Also per reference 4, the maximum rear axle weight for the loading and transport of the subject LMTV series vehicles only is 15,000 lbs.

d. GPK may need to be removed to meet C-17/C-5 height limitations depending on load plan.

e. C-17 & C-5: Do not use “Air Transport” or other lower tire inflation pressure settings. Tires must be inflated at highway/operational inflation pressure.

f. All hazardous materials (to include fuel level, batteries, etc.) must be prepared and certified for airlift in accordance with TM 38-250/AFMAN 24-204(I). Do not consider this air transport certification as approval for hazardous materials. Authorization for airlifting hazardous material is the responsibility of AFMC/A4RT (DSN 787-4503 or COM (937) 257-4503).

**4. Loading Instructions:**

a. C-130:

1) All certified vehicles can be loaded either front or aft end first into the aircraft with the exception of the M1086A1P2 Long Wheel Base with MHE (Material Handling Equipment – crane). This vehicle shall only be driven in front end first into the cargo compartment.

2) All FMTV variants: Axles shall be parked between FS 337 – FS 682 (LS 567 – LS 832 on C-130J-30).

3) Sleeper Shoring: Required at all times for C-130 aircraft.

**CAUTION:** Clearances as close as approximately 1.5” with respect to the vehicle cab area can exist as the vehicle traverses the C-130 ramp crest.

b. C-5 and C-17: Vehicles with height greater than 113” (e.g., GPK equipped trucks MTV LHS):

1) Sleeper Shoring: When required per paragraph 1b.

**CAUTION:** When loading the M1087 series vehicles; on C-5 aircraft; with the shelter on the back of the vehicle it is recommended to load them driven in through the forward loading complex only. If the vehicle must be backed in or loaded through the aft loading complex the loadmaster must use the projection charts in the 1C-5A-9 to determine if shoring is required.

2) C-5: Side-by-side loading may not be possible.
3) C-17: Maximum height for transport under or forward of aircraft wingbox is 150”.

**5. Restraint Requirements:** Any FMTV and all accompanying cargo must be restrained to meet air transport requirements of 3G forward, 1.5G aft and lateral, and 2G up. In addition, stored or installed equipment must meet these requirements and be capable of withstanding a 4.5G down load. Restraint should be applied using the four standard tiedown provisions, two at each end of the vehicle. If the tiedown provisions are not available or additional restraint is required, chains can be routed around solid frame points of the vehicle. Additional restraint can also be obtained from the vehicle axles up to their rated capacity (see Item Description paragraph), not to exceed 50% of the required restraint in longitudinal or lateral directions. Note that spring mounted axles provide zero vertical restraint capability. Cargo bed tiedown provisions are rated at 5,000 lbs each for use with secondary cargo.

**CAUTION:** The FMTV A1P2 tiedown front tiedown locations may have a shackle mounted in the small holes forward of the tiedown provision. Only use the large opening shown in Figure 3.



**Figure 3: Correct Tiedown Opening**

**Table 4: Restraint Capacity**

| Provision Location | Longitudinal | Lateral | Vertical |
|---|---|---|---|
| Forward | 92,967 lbs | 36,829 lbs | 24,860 lbs |
| Aft | 92,967 lbs | 33,279 lbs | 24,129 lbs |

**6. Minimum Sleeper Shoring Dimensions:**

a. C-130 (All Times): Four stacks (LMTV) or six stacks (MTV) at least: 23” L x 5.75” W (nominal 2x6 boards). The sleeper shoring is built up so that it is snug against the chassis/frame area and secured with cargo straps.

b. C-5 and C-17: Axles exceeding 19,000 lbs – 10” L x 23” W (2 stacks). The sleeper shoring is built up so that it is snug against the chassis/frame area and secured with cargo straps.



**Figure 4: Sleeper Shoring Placement**

**Required Distribution:**

1. Shipper shall give a copy of this certification to the ATOC representative when the item is presented for airlift. This certification shall be part of the official cargo manifest documentation package and shall be briefed to the aircraft loadmaster prior to loading these items.

2. AMC/A3V & AMC/A4T

3. SDDC TEA

**Point of Contact:** Tom McPeak, at thomas.mcpeak.1@us.af.mil or ATTLA@us.af.mil, DSN 986-9903 or Commercial (937) 656-9903. Refer to file number 2008.01.13 Rev 6 to reference this item.

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